

Requirement Analysis

FitBuddy AI — Project Deliverable

Translate the FitBuddy AI idea into user flow, data flow, functional requirements, and technology requirements.

Deliverables in this single phase document

- Customer Journey Map
- Data Flow
- Solution Requirements
- Technology Stack

Evidence status — Documented in the supplied README and project documentation.

2.1 Customer Journey Map

- 1. Open the FitBuddy AI home page.
- 2. Enter User ID, Name, Age, Weight, Goal, and Preferred intensity.
- 3. Select Generate My Plan.
- 4. Review the 7-day plan and Nutrition & Recovery Tip.
- 5. If required, submit feedback such as changing the warm-up time.
- 6. Review the revised plan.
- 7. For administration/testing, use the documented API or all-users view with appropriate protection.

2.2 Data Flow

Browser form → FastAPI route → request validation → plan generation logic / Gemini when enabled → SQLite persistence → result page. Feedback follows the same application flow to revise the plan and display the updated result.

Stage	FitBuddy AI behavior
Input	Profile fields, goal, preferred intensity
Validation	Pydantic / pydantic-settings configuration and input validation
Generation	Gemini integration when enabled; local demo mode is also supported
Persistence	SQLite through SQLAlchemy
Output	7-day activity plan + nutrition/recovery tip
Revision	Feedback submitted to revise the plan

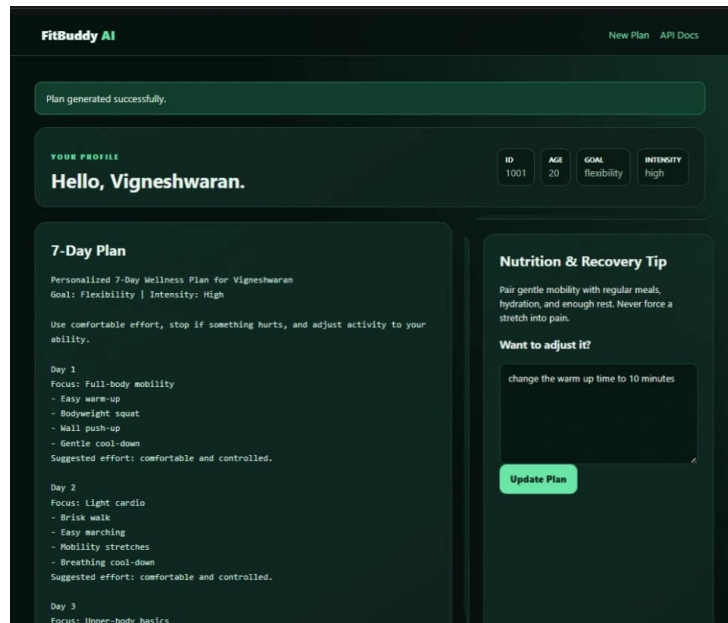
2.3 Functional & Non-Functional Requirements

- Functional: profile entry, 7-day plan generation, nutrition/recovery tip, feedback revision, JSON API endpoints, and all-users/admin view.
- Non-functional: responsive UI, input validation, automated tests, configurable model settings, and safe handling of secrets.
- Safety requirement: the application is a general-wellness demonstration and does not diagnose conditions or prescribe treatment.

2.4 Technology Stack

The documented stack includes Python, FastAPI + Uvicorn, Jinja2, SQLite + SQLAlchemy, Pydantic/pydantic-settings, Google GenAI SDK, pytest + httpx, and HTML/CSS.

Project Evidence / Screenshots



Screenshot evidence: generated 7-day plan and Nutrition & Recovery Tip.

Source note: This phase document was converted from the supplied FitBuddy AI project documentation, README, configuration/dependency information, and supplied project screenshots. Where the source did not provide a measured result, named team member, or actual date, the document does not present one as a completed fact.